UNIVERSITY of KIGALI "Unequalled Education Excellence"

CONTINUOUS ASSESSMENT TEST	
SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY	
PROGRAMME Bachelor degree with honours in computer science, Bachelor degree with honours in information technology and Bachelor degree with honours in business information technology	
MODULE CODE AND TITLE: **CSC83101: MACHINE LEARNING**	LEVEL 3, TRIMESTER 3
DATE: **22-02-2026**	**TIME:** 1:00PM-4:00PM **SESSION:** WEEKEND
DURATION: **3 HOURS**	**TOTAL MARKS:** 40
EXAMINER/ LECTURER: MR. TUYISHIMIRE JOSEPH	**MODERATOR(S):** Dr. Nteziriza Nkera Bahizi Josbert & Mr. TWESIGYE Geofrey

INSTRUCTIONS TO THE CANDIDATE

1. All questions must be attempted in the answer booklet provided.
2. All queries should be directed to the invigilator; do not communicate or attempt to communicate with any other candidate.
3. You have **THREE HOURS** to complete this paper. You are not allowed to leave the examination room within the **first hour** and in the last 15 minutes of this examination.
4. This is a **CLOSED BOOK** examination.
5. Answer 4 Questions in all, Question One in each section is Compulsory.
6. Choose any 1 other question in from Section A.
7. Choose any 1 other question in from Section B.
8. Please, do not write on the question paper.

SECTION A

QUESTION 1 (COMPULSORY) (10 marks)

At a mid-sized business, you work as a data analyst with the goal of improving workforce management and employee satisfaction. A thorough dataset comprising employee demographics, including age and gender, department, salary, and employee satisfaction ratings, has been collected by the organization. In order to guarantee accurate insights, this dataset also contains some missing values and inconsistencies that need to be fixed. In order to find patterns in employee satisfaction across different departments and demographic groups, it is your responsibility to perform an exploratory data analysis on this employee data. This entails calculating important summary statistics, such as the average salary, spotting any irregularities or issues with the quality of the data, and displaying the connections between satisfaction, age, and salary. HR strategies to increase employee engagement, retention, and departmental effectiveness will be guided by your

Age	Salary	Department	Gender
25	50000	HR	Male
30	60000	Finance	Female
NaN	55000	None	Female
22	NaN	IT	None
28	52000	Finance	Male
NaN	58000	HR	Female

a) Display the first six rows of the employee dataset to get an initial view of the data structure. This helps reveal the different columns and the types of information they contain, such as employee demographics, salary, department, and satisfaction scores. By examining these first few rows, you can quickly spot any missing values, inconsistencies, or unusual entries that may need to be addressed before further analysis. This step is crucial for gaining a clear understanding of the dataset's overall quality and organization, which informs how to proceed with cleaning and analyzing the data effectively. **(3 marks)**

b) Calculate and print the average salary from the employee dataset. Make sure to handle any missing or invalid salary entries appropriately so that the average reflects only valid salary data and gives an accurate result. This involves converting salary values to a proper numeric format, identifying and excluding any non-numeric or missing entries, and then computing the mean salary from the cleaned data. **(3 marks)**

c) You have a dataset containing employee information including Age and Gender. Since the dataset does not include a Satisfaction Score column, first generate this column by assigning random integer satisfaction scores between 1 and 10 for each employee. Then, write Python code to create a scatter plot showing Age versus Satisfaction Score, where each point is colored by gender, and use descriptive statistics to summarize the relationship between Age and Satisfaction Score. **(4 marks)**

QUESTION 2 (10 marks)

Your company's customer data currently suffers from incomplete and inconsistent entries, which are undermining the accuracy of your analyses and the performance of your machine learning models. To enhance data quality, a thorough cleaning process is needed. This begins with identifying and addressing missing values using suitable imputation techniques such as mean, median, or predictive methods to fill in the gaps. Inconsistent entries, particularly in categorical fields, must be standardized to maintain consistency throughout the dataset. Numeric fields should be examined for outliers, which need to be carefully handled either removed or transformed to avoid biased results and distortions during model training. Additionally, since many machine learning algorithms require numerical input, categorical variables should be converted into numeric formats via encoding methods like one-hot encoding or label encoding. Implementing this comprehensive cleaning approach will improve the accuracy of your analyses and boost the effectiveness of your predictive models, leading to better business decision-making.

Age	Gender	Annual_Income	Purchase_Amount	Customer_Segment	VisitsPer Month	Satisfaction_Score
59	F	63272.0	270	Medium	1	4
48	F	58041.0	292	Medium	3	6
56	M	56209.0	283	Medium	2	8
57	F	48738.0	352	Medium	2	10
59	M	NaN	274	Medium	5	3

a) Implement Python code to identify and print the count of missing values in each Column of the dataset. This task involves scanning the entire dataset to find how many entries are missing per feature, which is crucial for understanding the data quality and determining which columns require cleaning or imputation. **(4marks)**

b) Implement Python code to fill the missing values in the Annual Income column with the mean value of that column. This step addresses the incomplete data by replacing missing income values with the average income, thereby preserving the overall distribution and enabling downstream analyses without bias from missing data. **(3marks)**

c) Implement Python code to detect outliers in the Purchase_ Amount column using the Interquartile Range (IQR) method. Once detected, replace any outliers with the nearest boundary values defined by the lower and upper whiskers of the IQR range. This approach helps to mitigate the impact of extreme purchase amounts on statistical analyses and modeling, ensuring more robust results. **(3marks)**

QUESTION 3 (10 marks)

Your company is focused on enhancing its marketing strategies by gaining deeper insights into customer behavior and preferences. To achieve this, the goal is to identify natural groupings or segments within the customer base based on key attributes such as annual income, purchase amounts, and the frequency of visits each customer makes per month. By applying K-Means clustering, a widely used unsupervised machine learning technique, you can partition the customers into distinct clusters where members within each cluster share similar characteristics. This segmentation enables the business to better understand different customer profiles, tailor marketing campaigns to specific groups, optimize resource allocation, and ultimately improve customer engagement and sales effectiveness. After clustering, analyzing the distribution and characteristics of each cluster will provide actionable insights that support data-driven decision-making and targeted marketing efforts.

```
import pandas as pd
import numpy as np
np.random.seed(42)
n = 1000
data = {
    'Years_of_Experience': np.random.normal(loc=5, scale=2, size=n).astype(float),
    'Monthly_Sales': np.random.normal(loc=50000, scale=15000, size=n).astype(int),
    'Projects_Completed': np.random.poisson(lam=4, size=n)
}
df = pd.DataFrame(data)
# Introduce missing values in 'Years_of_Experience'
missing_indices = np.random.choice(df.index, size=50, replace=False)
df.loc[missing_indices, 'Years_of_Experience'] = np.nan
# Fill missing values with mean
mean_exp = df['Years_of_Experience'].mean()
df['Years_of_Experience'].fillna(mean_exp, inplace=True)
# Introduce outliers in 'Monthly_Sales' by multiplying some values by 5
outlier_indices = np.random.choice(df.index, size=10, replace=False)
df.loc[outlier_indices, 'Monthly_Sales'] = df.loc[outlier_indices, 'Monthly_Sales'] * 5
print(df.head())
```

a) Using the provided employee dataset, apply the K-Means clustering algorithm to segment the employees into three distinct clusters based on their years of experience, monthly sales, and number of projects completed. Properly initialize the model, fit it to the data, and assign each employee to one of the three clusters. Explain the parameters you choose for the model and briefly describe how K-Means works in this context. **(4marks)**

b) After clustering, add the resulting cluster labels as a new column in the original DataFrame. Calculate and display the number of employees that belong to each cluster to understand the distribution and size of each segment. Present the counts clearly and comment on any notable insights **(3marks)**

c) Create a two-dimensional scatter plot visualizing the clusters formed by K-Means clustering on the employee dataset. Plot YearsofExperience on the x-axis and Monthly_Sales on the y-axis. Use different colors or markers to distinguish the clusters visually. Include appropriate axis labels, a legend indicating cluster membership, and a descriptive title for the plot. This visualization will help interpret how clusters are distributed based on experience and sales. **(3marks)**

SECTION B

QUESTION 4 (COMPULSORY) (10 marks)

Your marketing team is aiming to enhance their customer targeting by accurately identifying which customers belong to the 'High' spending segment. To achieve this, they plan to leverage a combination of demographic and behavioral data points—including age, annual income, purchase amount, monthly visit frequency, and customer satisfaction scores. These features offer valuable insights into customer profiles and buying behaviors, enabling more precise segmentation. The dataset at hand has been carefully generated and preprocessed to reflect these variables, with a specific binary target variable created to indicate whether a customer is classified as a high spender or not. Additionally, the data has been split into training and testing subsets to facilitate model development and unbiased evaluation. Your role involves preparing this dataset for analysis, training a logistic regression classifier to predict high spending customers, and rigorously assessing the model's predictive performance. This process will support the marketing team in designing more effective, data-driven campaigns tailored to high-value customers, ultimately driving better business outcomes.

dex	YearsExperience	Salary
0	1.1	39343
1	2.0	46205
2	3.2	*NaN*
3	4.5	60000
4	*NaN*	65200
5	6.8	72500
6	7.5	*NaN*
7	8.3	83000
8	9.0	88000
9	10.5	95000

a) Train a simple linear regression model on the training data by initializing the regressor and fitting it with the training feature and target values. Simple linear regression is appropriate here because it models the relationship between one independent variable and a continuous dependent variable, allowing you to predict the target based on the given feature

(6marks)

b) Use the trained simple linear regression model to predict values on the test set and then compute the mean squared error (MSE) or another appropriate regression metric. This metric measures the average squared difference between the predicted and actual values, providing insight into the model's prediction accuracy. **(2marks)**

c) Extract and print the coefficient (slope) and intercept of the simple linear regression model. Interpret the sign of the coefficient: a positive coefficient means that as the independent variable increases, the target variable tends to increase, whereas a negative coefficient indicates an inverse relationship. The magnitude of the coefficient shows the strength of this effect. **(2marks)**

QUESTION 5 (10 marks)

Your company is developing a predictive model to classify customers into two groups based on behavioral and demographic features collected from recent market research. The dataset contains three key features capturing customer behavior patterns, along with a binary target indicating customer segment membership. To ensure robust model performance, the data has been preprocessed, including feature scaling. Your task is to build a Support Vector Machine (SVM) classifier that can accurately distinguish between the two customer groups using the prepared training data. After training, you will predict customer segments on unseen test data and evaluate the model's effectiveness using accuracy and confusion matrix metrics. This classification model will support targeted marketing efforts by identifying distinct customer groups based on their behavioral characteristics.

```
import pandas as pd

import numpy as np

np.random.seed(42)

n = 1000

Data = {

  'Task_Completion_Rate': np.random.normal(7, 2, n),

  'Customer_Satisfaction': np.random.uniform(1, 10, n),

  'Hours_Worked_Per_Week': np.random.normal(40, 5, n),

  'Performance': np.random.choice([0, 1], n, p=[0.5, 0.5])

}

df = pd.DataFrame(data)
```

a) Train a Support Vector Machine (SVM) classifier on the training data by initializing the model with appropriate parameters and fitting it to the scaled training features and corresponding performance labels. Use descriptive statistics (.describe()) to summarize the training features before and after scaling **(3marks)**

b) Use the trained SVM model to predict performance categories on the scaled test set. Use descriptive statistics to summarize the predicted labels distribution **(3marks)**

c) Evaluate the model's performance by calculating the accuracy score and generating the confusion matrix. Use appropriate summaries to interpret these metrics **(4marks)**

<u>QUESTION 6</u> (10 marks)

Your company is developing a predictive model to classify customers into two groups based on behavioral and demographic features collected from recent market research. The dataset contains three key features capturing customer behavior patterns, along with a binary target indicating customer segment membership. To ensure robust model performance, the data has been preprocessed, including feature scaling. Your task is to build a Decision Tree classifier that can accurately distinguish between the two customer groups using the prepared training data. After

training, you will predict customer segments on unseen test data and evaluate the model's effectiveness using accuracy and confusion matrix metrics. This classification model will support targeted marketing efforts by identifying distinct customer groups based on their behavioral characteristics

```
import pandas as pd

import numpy as np

np.random.seed(42)

n = 1000

data = {

  'Work_Hours_Per_Week': np.random.normal(40, 5, n),

  'Team_Interactions_Per_Day': np.random.normal(10, 3, n),

  'Task_Completion_Rate': np.random.normal(75, 15, n),

  'Engagement_Level': np.random.choice([0, 1], n, p=[0.7, 0.3])

}

df = pd.DataFrame(data)

print(df.head())
```

a) Train a Decision Tree classifier on the provided dataset by initializing the model with appropriate parameters and fitting it to the features and corresponding engagement labels. Explain why Decision Trees are well-suited for classification tasks and whether feature scaling is required for this algorithm **(4marks)**

b) Use the trained Decision Tree model to predict the engagement levels on a test set (split the dataset appropriately). This step evaluates how well the model generalizes to unseen employee data **(3marks)**

c) Evaluate the model's performance by calculating the accuracy score and generating the confusion matrix. Describe what these metrics indicate about the classifier's performance, highlighting its strengths and limitations. **(3marks)**

END